



Adama Science and Technology University
College of Electrical Engineering and Computing
Department of Software Engineering
Introduction to Systems Engineering (SEng4303)

Group Project Documentation

Title: Systems Engineering Approach to Developing a Hospital Information Management System (HIMS)

Submitted To: MR HUNDAOL BEDADA
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GROUP MEMBERS:**ID:**

Naol Endale Gemachu	UGR/25582/14
Negasa Reta Tefera	UGR/31072/15
Newal Hussien Yusuf	UGR/25353/14
Rabira Asefa Shifera	UGR/31113/15
Ruth Zewdu Abera	UGR/25730/14
Saleamlak Wendmnew Mekuria	UGR/25292/14
Salahadin Nuredin Husen	UGR/31168/15
Samrawit Alemu Abshiro	UGR/25320/14
Samson Tadesse Chernet	UGR/22578/13
Segni Merga Gutu	UGR/31202/15
Sewasew Tadele Yirsaw	UGR/25379/14
Sofoniyas Kedir Yesuf	UGR/25339/14
Sonte Adamu Wakoya	UGR/25780/14

1. Introduction

Modern healthcare systems are complex socio-technical environments that require accurate, timely, secure, and integrated information flows. Hospitals manage vast amounts of data related to patients, clinical procedures, staff, equipment, finances, and regulatory compliance. Traditional manual or fragmented digital systems lead to inefficiencies, medical errors, delayed decision-making, and increased operational costs.

A Hospital Information Management System (HIMS) is a comprehensive information system designed to integrate all hospital functions into a unified platform. Applying a Systems Engineering (SE) approach ensures that the system is developed in a structured, disciplined, and traceable manner throughout its life cycle.

This documentation provides an advanced, end-to-end Systems Engineering Life Cycle (SELC) application for developing a Hospital Information Management System, addressing technical, organizational, operational, and managerial aspects.

2. Project Purpose and Scope

2.1 Project Purpose

The primary purpose of this project is to:

- ✓ Apply Systems Engineering principles to a real-world healthcare system
- ✓ Design a robust, scalable, secure, and user-centered HIMS
- ✓ Demonstrate traceability from stakeholder needs to system implementation
- ✓ Reduce operational inefficiencies and improve patient care quality

2.2 Project Objectives

- ✚ Integrate clinical, administrative, and financial hospital processes
- ✚ Improve data accuracy, availability, and confidentiality
- ✚ Support clinical decision-making
- ✚ Ensure regulatory compliance
- ✚ Enable future system evolution

2.3 Project Scope

Included in Scope:

- ❖ Patient administration and medical records
- ❖ Appointment and scheduling management
- ❖ Laboratory, pharmacy, and billing modules

- ❖ Staff and resource management
- ❖ Reporting and analytics
- ❖ Excluded from Scope:
- ❖ AI-based diagnosis systems
- ❖ National-level health data exchange
- ❖ Medical device control systems

2.4 Assumptions

- Hospital has basic IT infrastructure
- Users have minimal computer literacy
- System is initially deployed in a single hospital

3. System Engineering Life Cycle Framework

Phase 1: Problem Definition & Stakeholder Analysis

3.1 Problem Definition

Hospitals face the following systemic problems:

- ✚ Disconnected departmental systems
- ✚ Paper-based records prone to loss and errors
- ✚ Long patient waiting times
- ✚ Inefficient billing and reporting
- ✚ Poor data visibility for management

These problems highlight the need for an integrated system that treats the hospital as a system-of-systems.

3.2 System Context

The HIMS operates within a broader healthcare ecosystem involving patients, healthcare professionals, regulatory bodies, insurance providers, and IT infrastructure.

3.3 Stakeholder Identification and Classification

Stakeholder Category Expectations

- Patients External Privacy, fast service
- Doctors Primary Users Reliable medical data
- Nurses Primary Users Workflow efficiency
- Hospital Management Decision Makers Reports and analytics
- IT Department Support Maintainability

- Government Regulator Compliance

3.4 Stakeholder Needs Analysis

- ✓ Functional accuracy
- ✓ High availability and reliability
- ✓ Ease of use
- ✓ Data security and privacy
- ✓ Regulatory compliance

Phase 2: Requirements Engineering

3.5 Requirements Development Process

- Stakeholder interviews
- Observation of hospital workflows
- Review of healthcare regulations
- Analysis of existing systems

3.6 Functional Requirements

FR-01: The system shall support patient registration with unique identifiers

FR-02: The system shall maintain complete electronic medical records

FR-03: The system shall support appointment scheduling and reminders

FR-04: The system shall manage laboratory test requests and results

FR-05: The system shall generate bills and manage payments

FR-06: The system shall support role-based access control

3.7 Non-Functional Requirements

- ✚ Security: Encryption, audit logs, access control
- ✚ Performance: Response time under peak load
- ✚ Reliability: Fault tolerance and backup
- ✚ Usability: Simple, intuitive user interface
- ✚ Maintainability: Modular design

3.8 Regulatory and Standards Requirements

- Health data protection laws
- ISO/IEC 25010 (Software Quality)

- ISO/IEC 27001 (Information Security)

3.9 Requirements Traceability

Traceability ensures each requirement maps to stakeholder needs and later to design, implementation, and testing.

Phase 3: Conceptual Design

3.10 System Concept of Operations (CONOPS)

The HIMS supports end-to-end hospital operations from patient entry to discharge, billing, and reporting.

3.11 Conceptual Architecture

- User Interface Layer
- Application Services Layer
- Data Management Layer

3.12 Subsystem Decomposition

- ✚ Patient Management Subsystem
- ✚ Clinical Services Subsystem
- ✚ Administrative Subsystem
- ✚ Financial Subsystem
- ✚ Reporting and Decision Support Subsystem

Phase 4: Preliminary Design

3.13 Logical Architecture Design

Defines interactions among subsystems using data flow and control logic.

3.14 High-Level Data Design

Key entities:

- Patient
- Medical Record
- Appointment
- Staff
- Billing

3.15 Interface Control Design

- ✓ User interfaces
- ✓ External interfaces
- ✓ Internal APIs

Phase 5: Detailed Design and Development

3.16 Detailed Architecture Design

1. Database normalization
2. API definitions
3. Authentication and authorization mechanisms

3.17 Technology Selection

1. Frontend: HTML, CSS, JavaScript
2. Backend: PHP / Java / Python
3. Database: MySQL / PostgreSQL
4. Security: SSL, hashing, access control

3.18 Development Strategy

- Iterative and incremental development
- Version control
- Continuous integration

Phase 6: Implementation (Construction/Production)

Sample Prototype

3.19 Prototype Features

- ✚ Secure login
- ✚ Patient registration
- ✚ Appointment scheduling
- ✚ Billing module
- ✚ Basic reporting

3.20 Deployment Strategy

- ✓ Pilot deployment
- ✓ Gradual rollout
- ✓ Training sessions

4. Integration, Testing, and Evaluation

Integration, testing, and evaluation are critical Systems Engineering activities that ensure the Hospital Information Management System (HIMS) satisfies all defined requirements and performs reliably in the real hospital environment.

4.1 Integration Strategy

Integration is performed incrementally to reduce risk and isolate defects early.

Integration Levels:

- Module-Level Integration: Individual modules (patient, billing, laboratory) are integrated with their internal components.
- Subsystem Integration: Related modules are combined (e.g., clinical subsystems).
- System Integration: All subsystems are integrated into a complete HIMS.
- External Integration: Interfaces with external systems such as insurance providers or government reporting systems.

Integration Principles

- ❖ Defined interface control documents (ICDs)
- ❖ Use of standardized data formats
- ❖ Continuous integration where possible

4.2 Verification and Validation (V&V)

- Verification: Ensures the system is built correctly according to specifications.
- Validation: Ensures the right system is built to meet stakeholder needs.
- Verification answers: "Did we build the system right?"
- Validation answers: "Did we build the right system?"

4.3 Testing Levels

Unit Testing: Tests individual functions or classes.

Integration Testing: Tests interactions between modules.

System Testing: Tests the complete system against requirements.

Performance Testing: Measures response time, load handling, and scalability.

Security Testing: Tests authentication, authorization, and data protection.

User Acceptance Testing (UAT): Conducted by doctors, nurses, and administrators.

4.4 Test Planning and Documentation

1. Test plans
2. Test cases
3. Test scripts
4. Test reports

4.5 Evaluation Metrics

Metric Description

- ❖ Accuracy Correctness of medical data
- ❖ Availability System uptime
- ❖ Response Time Speed of operations
- ❖ User Satisfaction Feedback from users

5. Operation, Support, and FRACAS

5.1 Operational Concept

1. Once deployed, HIMS operates continuously as a mission-critical system.
2. Operational Characteristics:
3. 24/7 availability
4. Concurrent multi-user access
5. Real-time updates

5.2 Operational Procedures

- User login and authentication
- Daily data backup
- Monitoring system performance
- Incident reporting

5.3 Support Structure

Level 1 Support: User issues and basic troubleshooting

Level 2 Support: Technical problem resolution

Level 3 Support: System enhancements and major fixes

5.4 Training and Documentation

- ✚ User manuals
- ✚ Administrator guides
- ✚ Training workshops

5.5 FRACAS (Failure Reporting, Analysis, and Corrective Action System)

FRACAS ensures continuous reliability improvement.

FRACAS Process Steps:

1. Failure detection
2. Failure reporting
3. Root cause analysis
4. Corrective action
5. Verification of fix
6. Preventive action

Example Failure Scenario:

- ✓ Failure: Billing module crash
- ✓ Cause: Database overload
- ✓ Correction: Query optimization

6. Retirement and Evolution Planning

6.1 System Retirement Planning

System retirement is planned early to avoid data loss and operational disruption.

Retirement Activities:

- ✚ Data archiving and migration
- ✚ Secure data deletion
- ✚ Hardware decommissioning

6.2 Legal and Compliance Considerations

- ❖ Medical data retention policies
- ❖ Patient privacy laws
- ❖ Audit requirements

6.3 Evolution and Upgrade Strategy

- ✚ Modular upgrades
- ✚ Feature expansion
- ✚ Integration with national health systems

6.4 Technology Refresh

- ✚ Database upgrades
- ✚ Security enhancements
- ✚ Infrastructure modernization

7. Systems Engineering Management

Systems Engineering Management ensures that technical and managerial activities are aligned throughout the project life cycle.

7.1 Project Planning

- Work Breakdown Structure (WBS)
- Schedule estimation
- Resource planning

7.2 Cost Management

Development cost estimation

Operation and maintenance costs

Lifecycle cost analysis

7.3 Risk Management

Risk Probability Impact Mitigation

Data breach Medium High Encryption, audits

User resistance Medium Medium Training

System downtime Low High Redundancy

7.4 Configuration Management

- Version control
- Baseline identification
- Change request management

7.5 Quality Management

- Quality standards
- Reviews and inspections
- Continuous improvement

7.6 Stakeholder Communication

- ✓ Progress reports
- ✓ Review meetings
- ✓ Feedback loops

8. Conclusion

This project presented a comprehensive application of the Systems Engineering approach to the development of a Hospital Information Management System (HIMS), demonstrating the value of a structured and lifecycle-oriented methodology for complex healthcare systems. By rigorously following the Systems Engineering Life Cycle (SELC)-from problem definition and stakeholder analysis through requirements engineering, design, implementation, integration, testing, operation, and retirement-the project ensured that stakeholder needs were systematically translated into verifiable requirements and robust technical solutions. The strong emphasis on verification and validation, integration, testing, and FRACAS highlights the importance of reliability, safety, and operational effectiveness in mission-critical hospital environments.

Overall, the project shows that successful development of a Hospital Information Management System is not only a technical task but also a managerial and organizational challenge. Through effective risk management, configuration control, quality assurance, and lifecycle planning, the proposed HIMS is designed to be secure, scalable, maintainable, and adaptable to future healthcare and technological changes. In conclusion, adopting a Systems Engineering approach provides a solid foundation for delivering sustainable, user-centered, and high-quality hospital information systems, while also serving as a strong academic and practical example of applying Systems Engineering principles to real-world healthcare challenges.